

Axel Tang

📍 Toronto, Canada 🌐 github.com/axeltwc  Axel Tang

ACADEMICS

University of Ottawa

2020 – 12/2023

Honours Bachelor of Science in Computer Science (Cum Laude)

- CGPA: 3.51/4.0
- Finished Early From Expected
- Honours and Awards: Dean's Honour List, International Merit Scholarship

PROFILE

- Software Developer • Passionate and Innovative
- Leadership Experience • Communicative and Visionary
- Teamwork Experience • Collaboration and Flexible
- Adaptive Schedules • Available In Office or Remote

SKILLS

Languages

Fluent: Java, Python, C#, HTML, CSS

Proficient: MATLAB, Racket-Scheme, Prolog, GoLang, JavaScript (React)

Quality Assurance

Fluent: Junit, Selenium

Proficient: JMH, EasyMocks, PitTest, Cucumber, JaCoCo, DBVisualizer

Database Management

Proficient: PostgreSQL, MongoDB, SpringBoot, Firestore, RESTful APIs, Postman

EXPERIENCE

Information Technology Intern

07/2024 – 08/2024

Hong Kong General Chamber Of Commerce 

- Implementing security cameras of office to have motion capture and face detection with **OpenCV Contours** using **Python**.
- Using **fuzzywuzzy algorithm** to compare data from **SQL database** to extract abbreviated names.
- Created Python program to extract IPs with **Regex** from input.
- Re-designing intranet webpages with **ASP.net** & **CSS** and data entry into system.
- Using **EsXi Host Client** for Virtual Machine Hosting and Maintenance.
- Restoring Data with Microsoft SQL Server Studio 2020.

Junior Software Developer Intern

03/2024 – 07/2024

InteractHealthPro 

- Transitioning from Keap to Groove.cm to enhance functionality and improve operational efficiency.
- Ensure seamless conversions while meeting and exceeding company standards.
- Collaborated with team leader and incorporating feedback to optimize migration process.
- Successfully migrated all assigned websites and email sequences with a 100% acceptance rate.

Open Source Project Developer

09/2023 – 12/2023

Umple - Open Source Project 

- Actively contributed to an open source project (model-oriented programming tool) with 2.41 million users.
- Tasked with implementing and resolving **Java** and **Umple** issues while ensuring **effective version control** and **CI/CD pipelining**.
- Submitted multiple **Pull Requests**; such as extraneous bracket detection with **Regex** and **Batch Scripts** for Windows Development
- Improved code quality and streamlined future development workflow.

HumblexMC Server

- Creating and ensuring server functionality, reliability and overall growth.
- Developed server plugins, automate deployment, establish online presence, lead the development team and enhance user engagement.
- Developed **Java** plugins with **Maven Bukkit API**, automated deployment with **Batch Scripts**, optimized the server's website with **SEO** to improve pagerank, led the team using **Trello** and **Miro**, and executed advertising campaigns achieving 465k views.
- Increased system reliability by 90%, improved pagerank from 33 to 4, effectively managed the development team, and significantly boosted server growth and user engagement.

PROJECTS

RANSAC Algorithm

Algorithm Creation Project

- From autonomous vehicles navigation , vehicles equipped with **LiDAR** sensors captures detailed 3D representation (point clouds) containing crucial information of structures such as roads and buildings.
- Assigned to implement an algorithm that detects the three most dominant planar structures with **Java, Go Lang, Prolog, Scheme** and output with a 99% confidence level.

NLP and DL

Artificial Intelligence (AI) Project

- Perform an empirical classification study and document the results.
- Utilizing **Python** with **Jupyter Notebook** and packages such as **Spacy, Pandas** and **Numpy** to visualize data and creating the NLP pipeline.
- Obtained macro/micro averages and classification report data (precision, recall, f1-score, support) for the **logistic regression model** and **MLP classifier model**.

Information Retrieval System

IR System Project

- Implementing a Information Retrieval System based on vector space model with a collection of documents.
- Utilizing **Python** to create preprocessing, indexing, retrieval and ranking algorithms.

2D AimTrainer

Desktop C# Application Project

- Created an Aim Trainer software in **C#** to improve aiming skills in FPS games.
- Implemented a score counter and miss counter to keep track of player statistics.
- Reducing target icon size per second by 10% after every iteration to increase difficulty.

eHotelsWebApp

DataBase Project

- Creating Server and Client Web Application to keep track of the entire Hotel System using technologies such as node, nodemon, express, postgres, cors, postman.
- Assigned to create the employee page and room page for the entire application with **Express.js, PostgreSQL, Postman (REST API)**.
- Successfully presented the project with pages working as intended.

uODentist Website

FrontEnd HTML+CSS+JS Project

- Assigned to create a frontend project using **HTML, CSS, JS, Bootstrap 5**.
- Created and modified the website to a friendly user-interface and match the University of Ottawa's color schemes.
- Received positive feedbacks and leveraged HTML and CSS skills.

TechSupport Website

FrontEnd React Project

- Developed frontend website for TechSupport Business.
- Utilized **React** to develop UI for the website.
- Created Semantic Network, Heuristic Evaluations and Interactive Processes.

StudentCourseBookingApp

Mobile Development Application

- Creating a demo project to book courses for students in University of Ottawa.
- Using **Java** to add validations utilizing regex for coursecode inputs in Android Studio.
- Finalizing the entire application creating **UML diagram**.